



CS607-ARTIFICIAL INTELLIGENTS

(Solved Macq's)

LECTURE FROM (23 to 45)



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1. Soft-computing is naturally applied in machine learning applications
 - ▶ **True** PG # 205
 - ▶ **False**
2. Genetic algorithms have been employed in finding the optimal initial weights of neural networks.
 - ▶ **True** PG # 205
 - ▶ **False**
3. Which is/are clustering algorithm(s)
 - ▶ Self-organizing maps
 - ▶ k-means
 - ▶ Linear vector quantization
 - ▶ **All of the given** PG # 205
4. Any given learning problem is primarily composed of _____ things
 - ▶ 4
 - ▶ **3** PG # 164
 - ▶ 1
 - ▶ 2
5. Fuzzy logic is a subset of conventional (Boolean) logic.
 - ▶ **True**
 - ▶ **False** PG # 147
6. A square is used to represent a Fuzzy set.
 - ▶ **True**
 - ▶ **False** PG # 151
7. Which one is not step involved in the planning phase of Linear model for expert systems
 - ▶ Feasibility assessment
 - ▶ Resource allocation
 - ▶ Task phasing and scheduling
 - ▶ **None of the given** PG # 129
8. _____ reasoning is used when the facts of the case are likely to change after some time
 - ▶ **Non-Monotonic** PG # 103
 - ▶ **Common-sense**
 - ▶ **Analogical**
 - ▶ **Abductive**
9. A statement in conjunctive normal form (CNF) consists of _____
 - ▶ Ors of ANDs
 - ▶ ANDs
 - ▶ **ANDs of Ors.** PG # 107
 - ▶ Ors
10. An expert system may take _____ main roles, relative to the human expert.
 - ▶ **Two** PG # 113
 - ▶ **Three**
 - ▶ **Four**
 - ▶ **Five**
11. Backward chaining is an inference strategy that works backward from a _____ to a _____.
 - ▶ **Hypothesis, Proof** PG # 126

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- ▶ Proof, Hypothesis
- ▶ Hypothesis, Experiment
- ▶ Proof, Conclusion

12. Which one is the general stage of the expert system development life cycle:

- ▶ Feasibility study
- ▶ Rapid prototyping
- ▶ Alpha system (in-house verification)
- ▶ All of the given **PG # 129**

13. In CLIPS, the command to load file is:

- ▶ **CLIPS (load “filename.clp”)** **PG # 137**
- ▶ CLIPS (load “filename”)
- ▶ CLIPS (“filename.clp”)
- ▶ CLIPS (open “filename.clp”)

14. In Linear Model, a linear sequence of steps is applied repeatedly in an iterative fashion to develop the ExpertSystem.

- ▶ **True** **PG # 129**
- ▶ False

15. _____ is the bottleneck in the construction of expert system.

- ▶ Planning
- ▶ **Knowledge acquisition** **PG # 130**
- ▶ Knowledge Design
- ▶ Code

16. Select the category that does NOT belong to Elicitation methods.

- ▶ Direct methods
- ▶ Indirect methods
- ▶ Informal discussions
- ▶ **Formal discussions** **PG # 131**

17. Inference networks encode the knowledge of rules and _____.

- ▶ Facts
- ▶ **strategies** **PG # 132**
- ▶ conditions
- ▶ none of the given

18. A classical set is a container, which wholly includes or wholly excludes any given element.

- ▶ **True** **PG # 145**
- ▶ False

19. _____ is the process by which the fuzzy sets that represent the outputs of each rule are combined into a single fuzzy set.

- ▶ **Aggregation** **PG # 157**
- ▶ Fuzzification
- ▶ Implication
- ▶ None of the given

20. Aggregation only occurs once for each output variable, just after the fifth and final

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step, defuzzification.

▶ **True** PG # 157

▶ False

21. Machine learning is a prerequisite for any mature program of artificial intelligence

▶ **True** PG # 160

▶ False

22. Many machine learning systems are classifiers.

▶ **True** PG # 161

▶ False

23. Fuzzy inference systems (FIS) are associated with a number of names

▶ **True** PG # 153

▶ False

24. Hypothesis space uses the _____ of the attributes.

▶ **Conjunctions (AND)** PG # 168

▶ Disjunctions (OR)

▶ Negation (NOR)

▶ None of the given

25. A drawback of FIND-S is that, it assumes the consistency within the training set.

▶ **True** PG # 173

▶ False

26. The Candidate-Elimination algorithm represents the _____

▶ **Version Space** PG # 173

▶ Solution Space

▶ Elimination Space

▶ None of the given

27. The first step of FIND-S is to initialize h to the most specific hypothesis in _____:

$h = \langle \emptyset, \emptyset \rangle$

▶ **H PG # 172**

▶ I

▶ J

▶ K

28. The Entropy is 1 when the collection contains number of positive examples _____ to/than negative examples.

▶ **Equal** PG # 177

▶ **Greater**

▶ Less

▶ None of the given

29. Artificial Neural Networks is a new learning paradigm which takes its roots from _____ inspired approach to learning.

▶ Chemistry

▶ Physics

▶ **Biology** PG # 181

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▶ Mathematics

30. A single layer perceptron can not perform pattern classification on linearly separable patterns.

▶ True

▶ **False**

PG # 186

31. Each neuron in the hidden layer is responsible for a different_____.

▶ Layer

▶ Neuron

▶ None of the given

▶ **Line**

PG # 186

32. In ANNs, Training is the heart of learning, in which finding the best_____that covers most of the examples is the objective.

▶ **Hypothesis**

PG # 189

▶ Neuron

▶ Agent

▶ Operator

33. Action is a_____used to change states.

▶ **Predicate**

PG # 198

▶ Function

▶ Operator

▶ None of the given

34. Clustering is a form of_____learning.

▶ Supervised

▶ **Unsupervised**

PG # 205

▶ Guided

▶ Unguided

35. Clustering is a form of unsupervised learning.

▶ **True**

PG#205

▶ False

36. A concept is the representation of the problem with respect to the given attributes.

▶ **True**

PG#167

▶ False

45. Which one is not step involved in the planning phase of Linear model for expert systems

▶ Feasibility assessment

▶ Resource allocation

▶ Task phasing and scheduling

▶ **None of the given**

PG#129

46. Breadth-First Search checks all paths of a given length before moving on to any longer paths.

▶ **True**

▶ False

47. A statement in conjunctive normal form (CNF) consists of _____

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- ▶ Ors of ANDs
- ▶ ANDs

▶ **ANDs of Ors.** **PG # 107**

- ▶ Ors

48. The goal of an Assisting Expert is to aid an expert in a routine tasks to increase

- ▶ Planning
- ▶ Execution

▶ **Productivity** **PG # 114**

- ▶ Correctness

49. Which one is the general stage of the expert system development life cycle:

- ▶ Feasibility study
- ▶ Rapid prototyping
- ▶ Alpha system (in-house verification)

▶ **All of the given** **PG # 129**

50. The Linear model of software development has been successfully used in developing systems.

▶ **Expert** **PG # 129**

- ▶ Software
- ▶ Design
- ▶ Logical

51. The main phases of the linear sequence are:

- ▶ Knowledge acquisition and analysis
- ▶ Knowledge design
- ▶ Code

▶ **All of the given** **PG # 129**

52. The Defrule construct is used to _____ rules.

- ▶ define

▶ **add** **PG # 135**

- ▶ declare
- ▶ none of the given

53. The goal of knowledge analysis is to analyze and structure the ____ gained during the knowledgeacquisition phase.

▶ **Knowledge** **PG # 131**

- ▶ facts
- ▶ rules
- ▶ conclusions

54. Inference networks encode the knowledge of rules and _____.

- ▶ facts

▶ **strategies** **PG # 132**

- ▶ conditions
- ▶ none of the given

55. Fuzzy sets, unlike _____ sets, do not restrict themselves to something lying

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wholly in either set A or inset not-A.

▶ **Classical** PG # 146

- ▶ Physical
- ▶ Universal
- ▶ None of the given

56. Reasoning in fuzzy logic is just a matter of generalizing the familiar _____ logic.

▶ **Boolean** PG # 147

- ▶ Complex
- ▶ Coognitive
- ▶ Supervised

57. _____ logic lets us define more realistically the true functions that define real world scenarios.

▶ **Fuzzy** PG # 148

- ▶ Classical
- ▶ Boolean
- ▶ None of the given

▶ **Membership** PG # 149

- ▶ Ordinary
- ▶ Fuzzy
- ▶ Inline

58. Usually a _____ graph is chosen to represent a fuzzy set.

▶ **Triangular** PG # 151

- ▶ Circular
- ▶ Conical
- ▶ None of the given

59. In Fuzzy Rules there are two parts to the antecedent, and they have a/an _____ operator in between them.

- ▶ AND

▶ **OR** PG # 153

- ▶ NOT
- ▶ None of the given

60. If the antecedent is only partially true, then the output fuzzy set is truncated according to the _____ method.

- ▶ Intrinsic

▶ **Implication** PG # 153

- ▶ Boolean
- ▶ None of the given

61. The role of tester is often called the critic.

▶ **True** PG # 160

- ▶ False

62. Inductive learning takes examples and generalizes rather than starting with knowledge.

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► Existing PG # 162

- Inductive
- Deductive
- None of the given

63. The tractable problems are further divided into structured and _____ problems

- Non-structured

► Complex PG # 166

- Simple
- None of the given

64. Hypothesis space uses the _____ of the attributes.

► Conjunctions (AND) PG # 168

- Disjunctions (OR)
- Negation (NOR)
- None of the given

65. FIND-S finds the maximally specific hypothesis possible within the _____.

► Version space PG # 172

- Solution space
- Hypothesis space
- None of the given

66. Entropy characterizes the purity/impurity of an arbitrary collection of examples.

► True PG # 177 NOT SURE

- False

73. A single Perceptron simply draws a line, which is a hyper plane when the data is _____ than 2 dimensional.

► More PG # 184

- Less
- Equal
- None of the given

74. In ANNs, Training is the heart of learning, in which finding the best that covers most of the examples is the objective.

► Hypothesis PG # 189

- Neuron
- Agent

75. Operator In planning phase, each state is represented in predicate logic.

► True PG # 197

- False

76. Action is a _____ used to change states.

► Predicate PG # 198

- Function
- Operator
- None of the given

77. Which one is NOT the application area of Computer Vision?

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- ▶ Tracking an object through an image sequence
- ▶ Object Extraction from a video sequence
- ▶ Automated Navigation of a robot or a vehicle
- ▶ **None of the given** **PG # 203**

78. Naturally, there is no supervision of classification in _____ algorithms for their learning or clustering.

- ▶ **Clustering** **PG # 205**
- ▶ Binary
- ▶ Planning
- ▶ Searching

79. Which one is NOT the feature of Robot:

- ▶ Reasoning, Dealing with uncertainty
- ▶ Vision, Learning
- ▶ Autonomy, Physical Intelligence
- ▶ **None of the given** **PG # 204**

80. Intelligence is the ability to;

- Think /learn plan /schedule
- Recognize / remember
- Problem solving
- **All of the above**

81. Computer vision encompasses topic(s) from

- Image Processing
- Machine learning
- **All of the given** **PG # 203**
- Pattern recognition

82. In theoretical computer science there are two main branches of problems:

- ▶ **Tractable and intractable** **PG # 165**
- ▶ Intractable and induction
- ▶ Tractable and induction
- ▶ None of the given

83. In theoretical computer science there are _____ main branches of problems

- 1
- **2** **PG # 165 Tractable and Intractable**
- 3
- 4

84. There are _____ phases in machine learning.

- 1
- 2
- **3** **PG # 160 1. Training, 2. Validation, 3. Application**
- 4

85. Decision trees give us disjunctions of conjunctions, that is, they have the form:

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(A AND B) _____ (C AND D).

○ **OR** **PG # 176**

- AND
- XOR
- None of the given

86. Decision trees give us conjunctions of disjunctions.

▶ True

▶ **False PG # 176**

87. The input for the defuzzification process is a fuzzy set (the aggregate output fuzzy set) and the output is a double number.

▶ True

▶ **False PG # 157**

88. Which one is not step involved in the planning phase of Linear model expert systems

- ▶ Feasibility assessment
- ▶ Resource allocation
- ▶ Task phasing and scheduling

▶ **None of the given** **PG # 129**

89. The goal of knowledge analysis is to analyze and structure the knowledge gained during the planning phase.

▶ True

▶ **False** **Page # 131**

90. is an expert system which was developed at Stanford to aid physicians in diagnosing and treating patients with a particular blood disease

▶ **MYCIN** **Page # 112**

- ▶ DENDRAL
- ▶ R1/XCON
- ▶ R3/XCON

91. Another expert system named _____ was developed by Digital Equipment Corporation, as a computer configuration assistant.

▶ **R1/XCON** **Page # 112**

- ▶ MYCIN
- ▶ DENDRAL
- ▶ R3/XCON

92. The goal of an Assisting Expert is to aid an expert in a routine tasks to increase

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▶ Planning

▶ Execution

▶ **Productivity** PG # 114

▶ Correctness

93. Which one is the general stage of the expert system development life cycle:

▶ Feasibility study

▶ Rapid prototyping

▶ Alpha system (in-house verification)

▶ **All of the given** PG # 129

94. Select the category that does NOT belong to Elicitation methods.

▶ Direct methods

▶ Indirect methods

▶ Informal discussions

▶ **Formal discussions** PG # 131

95. A classical set is a container, which wholly includes or wholly excludes any given element.

▶ **TRUE** PG # 145

▶ **FALSE**

96. It was Aristotle who came up with the „Law of the Excluded Middle“.

▶ **True** PG # 145

▶ **False**

97. It was _____ who came up with the „Law of the Excluded Middle“.

▶ **Aristotle** PG # 145

▶ **Newton**

▶ **Einstein**

▶ **None of the given**

98. Reasoning in fuzzy logic is just a matter of generalizing the familiar _logic.

▶ **Boolean** PG # 147

▶ **Complex**

▶ **Coognitive**

▶ **Supervised**

99. If there are multiple parts to the antecedent, apply fuzzy logic _____ and resolve the antecedent to a single number between 0 and 1.

▶ **Operators** PG # 153

▶ **Rules**

▶ **Conditions**

▶ **None of the given**

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100. Outputs of learning are determined by the _____

▶ **Application** PG # 161

- ▶ Validation
- ▶ Training
- ▶ None of the given

101. Inductive learning is based on the knowledge that if something happens a lot it is likely to be generally _____

▶ **True** PG # 161

- ▶ False
- ▶ Ambiguous
- ▶ None of the given

102. _____ learning works on existing facts and knowledge and deduces new knowledge from the old.

▶ **Deductive** PG # 162

- ▶ Inductive
- ▶ Application
- ▶ None of the given

103. A concept is the representation of the _____ with respect to the given attributes.

- ▶ Solution

▶ **Problem** PG # 167

- ▶ Knowledge
- ▶ None of the given

104. Hypothesis space uses the _____ of the attributes.

▶ **Conjunctions (AND)** PG # 168

- ▶ Disjunctions (OR)
- ▶ Negation (NOR)
- ▶ None of the given

115. In all calculations involving Entropy we define _____ to be _____

▶ **$0 \log 0, 0$** PG # 177

- ▶ $0 \log 10, 1$
- ▶ $0 \log 0, 1$
- ▶ $1 \log 1, 1$

116. Measure of the effectiveness of an attribute in classifying the training data is called.

▶ **Information Gain** PG # 177

- ▶ Measure Gain

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- Information Goal
- None of the given

117. The soma and the enclosed nucleus in neuron play a significant role in the processing of incoming and outgoing data.

- True

➤ **False** PG # 181

119. A single layer perceptron cannot perform pattern classification on linearly separable patterns.

- True

▪ **False** PG # 186

120. In planning phase, each state is represented in predicate logic.

○ **True** PG # 197

- False

121. Action is a _____ used to change states.

○ **Predicate** PG # 198

- Function
- Operator
- None of the given

122. _____ is a subfield of _____.

➤ **Computer vision, Artificial Intelligence** PG # 203

- Robotics, Artificial Intelligence
- Soft computing, Artificial Intelligence
- None of these

123. _____ deals with procedures that extract useful information from static pictures and sequence of images.

➤ **Computer vision** PG # 203

- Neural networks
- Predicate logic
- None of the given

124. Which of the combinations is possible to solve real world problems?

- Genetic fuzzy
- Neuro –Fuzzy systems
- Neuro –Genetic systems

➤ **All of the given** PG # 205

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125. Genetic Algorithm applied on _____ problems?

- Real
- **All**
- Selected
- None

123. Answering the Sequence Problem (1, 3, 5, 11, ?) need

▶ **Intelligence**

- ▶ None of the given
- ▶ Sorting
- ▶ Searching

124. ___ reasoning is an informal form of reasoning that uses rules gained through experience or what we call rules-of-thumb

- Inductive
- Deductive
- Abductive

○ **Common-sense**

125. A _____ is "A person who possesses the skill and knowledge to solve a specific problem in a manner superior to others"

▶ **The domain expert**

Page # 122

- ▶ The knowledge engineer
- ▶ The end user
- ▶ All of the given

126. Best-first search always moves _____ from the node that seems closest to the goal node.

- Backward
- Left
- Right

○ **Forward**

127. Which one of the following is involved in an ES development project:

- ▶ The domain expert
- ▶ The knowledge engineer
- ▶ The end user

▶ **All of the given**

PG # 122

128. In backward chaining terminology, the hypothesis to prove is called the

- ▶
- ▶ Proof

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▶ Goal PG # 126

- ▶ Plan
- ▶ None of the given

129. ___ chaining is more focused and tries to avoid exploring unnecessary paths of reasoning.

- Forward

○ Backward PG # 128

- Both forward and backward
- None of the given

130. Assisting an expert is the most commonly found role of an Expert System.

- ▶ False

▶ True PG # 114

131. Choose the fields in which Fuzzy inference systems have been successfully applied:

- ▶ automatic control
- ▶ data classification
- ▶ decision analysis

▶ All of the given PG # 153

132. Mamdani's method was among the first _____ built using fuzzy set theory.

▶ control systems PG # 153

- ▶ expert system
- ▶ decision analysis system
- ▶ none of the given

133. Which one is NOT the phase of machine learning:

- ▶ Training
- ▶ Application
- ▶ Validation

▶ None of the given PG # 160

134. ___ is the process of formulating the mapping from a given input to an output using Fuzzy logic.

▶ FIS PG # 153

- ▶ FOS
- ▶ FIZ

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- ▶ None of these

135. Machine learning typically follows __phases according to Finlay.

- Two
- **Three PG # 160**
- Four
- Five

136. In context of tree, an arrow from one node to other is called :

- ▶ Root
- ▶ **Edge**
- ▶ Ancestor
- ▶ Descendant

137. Robotics have active contributions from _____

- ▶ Biology
- ▶ Psychology
- ▶ Mathematics
- ▶ **All of the given page # 204**

138. Identify the step involved in planning phase.

- Coding
- **Resource allocation page # 129**
- Knowledge acquisition from expert
- Identify concrete knowledge elements

139. Which of the following command is correct for adding numbers in CLIPS?

- ▶ CLIPS> (3 + 4)
- ▶ **CLIPS> (+ 3 4) page # 133**
- ▶ CLIPS> (3 4 +)
- ▶ CLIPS> (+ 3 4 +)

140. During __stage knowledge engineer works with the domain expert to acquire, organize and analyse the domain knowledge.

- ▶ System Design
- ▶ Feasibility Study
- ▶ System Planning
- ▶ **Knowledge acquisition page # 129, 130**

141. Which of the following command is used to see the added facts in CLIPS?

- ▶ **(facts) page 134**
- ▶ (fact)
- ▶ Fact

142. The correct command to load a CLIPS file is:

- (read "filename.clp")

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- (loadfile “filename.clp”)

- (load “filename.clp”) page # 137

- (readfile “filename.clp”)

143. In CLIPS, ___ command is used to remove facts.

- ▶ facts

- ▶ retract page # 134

- ▶ erase

- ▶ delete

144. The goal of ___ is to analyse and structure the knowledge gained during the knowledge acquisition phase.

- ▶ Knowledge structure

- ▶ Knowledge design

- ▶ Knowledge analysis page # 131

- ▶ ES design

145. In context of ES development, which of the following will be part of planning phase?

- ▶ Identify concrete knowledge elements

- ▶ Feasibility assessment page # 129

- ▶ Coding

- ▶ Knowledge acquisition from expert

146. In CLIPS, the ___ command is used for debugging programs.

- ▶ FACT

- ▶ CLEAR

- ▶ DEBUG

- ▶ WATCH page # 135

147. Knowledge elicitation methods may be classified into:

- ▶ Direct and indirect methods page #130,131

- ▶ Hierarchal and non hierarchal methods

- ▶ Data driven and application driven methods

- ▶ Direct and hierarchal

148. CLIPS stands for:

- ▶ C# Language Integrated Production System

- ▶ COBOL Language Integrated Production System

- ▶ C++ Linguist Integrated Production System

- ▶ C Language Integrated Production System page # 133

149. Identify the sequence wise main phases of Linear model used in developing expert system

- 1) Planning 2) Knowledge acquisition and analysis 3) Knowledge design 4) System evaluation 5) Code 6) Knowledge verification

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- 1)Planning 2)Knowledge acquisition and analysis 3)Knowledge design 4) Code 5)Knowledge verification 6) System evaluation page # 129
 - 1)Planning 2)Knowledge acquisition and analysis 3)Knowledge design 4) Code 5) System evaluation 6) Knowledge verification
 - 1)Planning 2)Knowledge acquisition and analysis 3)System evaluation 4) Knowledge design 5)Code 6)Knowledge verification
- 150.In context of CLIPS, “agenda” is the list of _
- ▶ Activated rules page # 136
 - ▶ Activated relations
 - ▶ Activated deftemplates
 - ▶ Deactivated rules
- 151._____encode the knowledge of rules and strategies.
- ▶ ES model
 - ▶ ES requirements
 - ▶ Knowledge base
 - ▶ Inference networks page # 132
- 152.Which of the following is one of the general stages of ESDLC?
- ▶ Spiral model
 - ▶ Design coding
 - ▶ Linear model
 - ▶ Beta system (tested by users) page # 129
- 153.Identify the correct step used to start design of an expert system.
- ▶ Scope study
 - ▶ Rapid prototyping
 - ▶ Feasibility study page # 129
 - ▶ Problem reorganization
- 154.Knowledge acquisition techniques may not include:
- ▶ Interviews
 - ▶ On-site observation
 - ▶ Electronic brainstorming
 - ▶ Surveys page # 130
- 155.Identify the correct definition of linear model given below.
- ▶ A non linear sequence of steps is applied repeatedly in an iterative fashion to develop the expert systems.
 - ▶ A non sequential set of steps is applied repeatedly in an iterative fashion to develop the expert systems.
 - ▶ A linear sequence of steps is applied repeatedly in an iterative fashion to develop the expert systems. Page # 129
 - ▶ A arbitrary set of steps is applied repeatedly in an iterative fashion to develop the software models.
- 156.If more than one domain experts have to be consulted for knowledge acquisition, than its better to use:

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- ▶ Surveys
- ▶ Electronic brainstorming
- ▶ Interviews

▶ **Blackboard system** page # 130

157. Identify the correct statement to list facts numbers 1 through 10

▶ clips> (facts 10)

▶ **clips> (facts 1 10)** page # 134

▶ clips> (facts 1)

▶ clips> (facts 1 to 10)

158. Reasoning in fuzzy logic is just a matter of generalizing the familiar ____ logic.

- ▶ Complex
- ▶ Supervised
- ▶ Cognitive

▶ **Boolean** page # 147

159. Problem faced during knowledge acquisition may not include

▶ Expert may provide incomplete knowledge

▶ **Expert provide exact and accurate knowledge** page 131

▶ Expert may not provide relevant information

▶ Expert may provide inconsistent or incorrect knowledge

160. Identify that for which purpose statement given below is used (deftemplate Person

```
(slot name
(type STRING
)) (slot
age(type
NUMBER))
```

- ▶ Attributes of a fact
- ▶ Rule definition

▶ **CLIPS** page # 134

▶ Rule

160. In CLIPS, ____ command erases contents of working memory

▶ (retract)

▶ (remove)

▶ (erase)

▶ **(clear)** page # 136

161. Identify the correct statement for the given rule

IF

IELTS score of

ali is 6 and

CGPA score of

Ali is 3.7 and

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GRE score of
ali is 66
Then

He is eligible to take admission in any of the university abroad

- ▶ (deftemplate Ali (slot attribute)(slot value))
(defrule Alichance (Ali (attribute "IELTS score")(value "6")) (Ali (attribute "CGPA")value "3.7"))
- ▶ (defrule Alichance (Ali (attribute "IELTS score")(value "6")) (Ali (attribute "CGPA")value "3.7"))
(Ali (attribute "GRE")value "66"))

Page # 135

- ▶ (deftemplate Ali (slot attribute)(slot value))
(defrule Alichance (Ali (attribute "IELTS score")(value "6")) (Ali (attribute "CGPA")value "3.7"))
- ▶ (deftemplate Ali (slot attribute)(slot value))
(defrule Alichance (Ali (attribute "IELTS score")(value "6")) (Ali (attribute "CGPA")value "3.7"))

162. Identify correct statement
for the given rule IF

The aptitude level of ungraduated
student is low and The English level
of ungraduated student is dull Then
he is not eligible to go abroad for
higher studies

- ▶ (deftemplate to UnderGradstudent (slot attribute)(slot values))
(defrule StudentStatus (UnderGradStudent(attribute "aptitudelevel")(value "low")) (UnderGradStudent(attribute "English understanding level")(value "dull"))

=>

(Printout " he is not eligible to go abroad for higher studies"))

- ▶ deftemplate to UnderGradstudent (slot attribute)(slot values))
(defrule StudentStatus (UnderGradStudent(attribute "aptitudelevel")(value "low")) (UnderGradStudent(attribute "English understanding level")(value "dull"))
<=>

(Printout " he is not eligible to go abroad for higher studies"))

- ▶ (defrule StudentStatus
(UnderGradStudent(attribute
"aptitudelevel")(value "low"))
(UnderGradStudent(attribute "English understanding level")(value "dull"))

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=>

(Printout “ he is not eligible to go abroad for higher studies”)) **Page # 135**

- ▶ deftamplate to UnderGradstudent (slot attribute)(slot values))
(defrule StudentStatus (UnderGradStudent(attribute
“aptitudelevel”)(value”low”)) (UnderGradStudent(attribute
“English understanding level”)(value “dull”))
=>
(Printout “ he is not eligible to go abroad for higher studies”))

163.In CLIPs the defrule construct is used to add

▶ **rules** **Page # 135**

- ▶ agenda
- ▶ facts
- ▶ principles

164. In the context of ES development which of the following will be part of planning phase

- ▶ knowledge acquisition from expert
- ▶ identify concrete knowledge element

▶ **feasibility assessment** **Page # 129**

- ▶ coding

164.identify the sets on which Membership Function is used

- ▶ Crisp set
- ▶ Classical set
- ▶ **Fuzzy set**
- ▶ None of Given

165. Fuzzy___ system have multidisciplinary nature.

- ▶ Intelligent
- ▶ Set
- ▶ Logic
- ▶ **Interface**

166.For computing more complicated functions we use ANN, which stand for?

- ▶ Artificial Next Network
- ▶ **Artificial Neural Network**
- ▶ Acid Neural Network
- ▶ Artificial Neuron Network

167.In unsupervised search a,_____represents some specific underlying pattern in the Data.

- ▶ **Cluster**
- ▶ Chunks
- ▶ Fragments
- ▶ Segments

168. Identify the correct definition of linear model given below.

- ▶ An arbitrary set of steps is applied repeatedly is an iterative fashion to develop the software models.
- ▶ **A linear sequence of steps is applied repeatedly is an iterative fashion to develop the**

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Expert System

- ▶ A Non-linear sequence of steps is applied repeatedly in an iterative fashion to develop the Expert System
- ▶ A Non sequence of steps is applied repeatedly in an iterative fashion to develop the Expert System

169. In ANNs, MSE is known as:

- ▶ Mean Sequential Error
- ▶ Medium Squared Error
- ▶ Most Sequential Error
- ▶ Most Spherical Error

170. Usually a ____ graph is chosen to represent a fuzzy set.

- ▶ Circular
- ▶ Conical
- ▶ Triangle
- ▶ Bar

171. Learning denotes changes in a system that enables a system to do the same task more-----
---the next time.

- ▶ Efficiency
- ▶ Conveniently
- ▶ Effectively
- ▶ With Reliability

172. ____ Learning does not work on existing facts.

- ▶ Deductive
- ▶ Exclusive
- ▶ Inductive
- ▶ Intelligent

173. Which one is the advantage of Neural Network

- ▶ Good for generalization (Page 187)
- ▶ The power of ANNs lies in their parallel architecture
- ▶ Less defined rules to build a neural network
- ▶ Knowledge implicit

174. If the true output of a concept $c(x_i)$ is 1 or 0 for an instance, then the output by our hypothesis $h(x_i)$ is 1 or 0 as well, respectively.

- ▶ True (Page 177)
- ▶ False

175. Which statement about learning is true:

- ▶ Learning is constructing or modifying representations of what is being experienced
- ▶ Learning denotes changes in a system that enables a system to do the same task more efficiently the next time.
- ▶ Learning is making useful changes in our minds.
- ▶ All of the given (Page 159)

176.

Please choose one _____ is a subfield of _____.

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- ▶ Computer vision, Artificial Intelligence (Page 203)
- ▶ Robotics, Artificial Intelligence
- ▶ Soft computing, Artificial Intelligence
- ▶ None of these

177.

Fuzzy logic is a superset of _____ logic.

- ▶ conventional (Page 147)
- ▶ positive
- ▶ negative
- ▶ None of these

178. Any statement can be fuzzy

- ▶ True (Page 147)
- ▶ False

179.

Within an expert system, the _____ contains facts about a specific subject area and rules that express the reasoning procedures of an expert on the subject.

- ▶ Inference engine
- ▶ Knowledge engineer
- ▶ Knowledge base
- ▶ None of the given options

180.

The brain is a collection of about 100 _____ interconnected neurons.

- ▶ Million
- ▶ Billion (Page 181)
- ▶ Trillion
- ▶ None of the given

181. _____ stands for interactive dichotomize

- ▶ ID (Page 177)
- ▶ IDC
- ▶ Int D
- ▶ None of the given

182. Complex problems usually have well-defined steps

- ▶ True
- ▶ False (Page 165)

183. the input of aggregation is the list of truncated output functions returned by the _____ process for each rule.

- ▶ truncation
- ▶ implication (Page 184)

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- ▶ Aggregation
- ▶ None of the given

184. Identify the sets in which Member function is used.

- ▶ Crisp set
- ▶ Classical set
- ▶ Fuzzy set (Page 149)
- ▶ None of the above

185. Identify the statement which best defines fuzzy sets.

- ▶ Fuzzy sets, unlike classical sets, restrict themselves to something lying wholly in either set A or in set not-A.
- ▶ Fuzzy sets, like classical sets, restrict themselves to something lying wholly in either set A or in set A.
- ▶ Fuzzy sets, unlike classical sets, do not restrict themselves to something lying wholly in either set A or in set not-A. (Page 146)
- ▶ Fuzzy sets, like classical sets, do not restrict themselves to

186. In Fuzzy Inputs we resolve all fuzzy statements in the antecedent to a degree of membership between 0 and __.

- ▶ 1 (Page 152)
- ▶ 2
- ▶ 3
- ▶ 4

187. The Multilayer Perceptron's are the most basic artificial neural _____

- ▶ Network (Page 186)
- ▶ System
- ▶ Interface
- ▶ None of these

188. Which of the following window is not present in CLIPS tool?

- ▶ Rule window Google
- ▶ Facts window
- ▶ Focus window
- ▶ Agenda window

189. Which of the following is correct CLIPS command to display the product of two numbers "5" and "3"?

- ▶ (Printout t "Product of two number is " (* 5 3)) Google
- ▶ (Printout t "Product of two number is " (x 5 3))
- ▶ (Printout t "Product of two number is " (5 x 3))
- ▶ (Printout t "Product of two number is " (5 * 3))

190. CLIPS command for multiplying two numbers 3 and 4 is

- ▶ CLIPS> *(3 4)
- ▶ CLIPS> (3 4*)

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▶ CLIPS> (3 * 4)

▶ CLIPS> (* 3 4)

Google

191. Identify which statement defines classical sets in best way

- ▶ A classical set is a container, which wholly includes but not wholly excludes any given
- ▶ A classical set is a container, which does not wholly include or wholly exclude any given element
- ▶ A classical set is a container, which sometimes wholly includes or wholly excludes any given element
- ▶ **A classical set is a container, which wholly includes or wholly excludes any given element** PG # 145

192. Problems faced during knowledge acquisition may include:

- ▶ Expert may not provide relevant information
- ▶ Expert may provide incomplete knowledge
- ▶ Expert may provide inconsistent or incorrect knowledge
- ▶ **All of the given options** PG # 131

193. The input of the aggregation process is the list of truncated output functions returned by the-----Process for each rule.

- ▶ Truncation
- ▶ **Implication** PG # 157
- ▶ Aggregation
- ▶ None of the given

193. The “ concept of partial truth ” can best be handled by:

- ▶ **Fuzzy logic** PG # 1
- ▶ Boolean logic
- ▶ Classical Logic
- ▶ Parametric logic

194. Knowledge elicitation method may be classified into:

- ▶ **Direct and Indirect Methods** Page # 130,131
- ▶ Hierarchical and No Hierarchical method
- ▶ Data Driven and Application Driven Methods
- ▶ None of Given option

195.

In fuzzy set and membership function, the input is always a _____ value.

- ▶ **Crisp Numerical** NOT SURE
- ▶ Fixed
- ▶ Amounts
- ▶ Interval Based

196.

The degree of truth is specially driven out by a function called the _____ function.

- ▶ **Ordinary**
- ▶ **Membership**

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- ▶ **Boolean**

- ▶ **Inline**

197. The Role of Taster is often called the _____ role.

- ▶ **Critical**

- ▶ **Real**

- ▶ **Soft**

- ▶ **Sophisticated**

198. Which of the Following is not The main phases of the linear sequence?

- ▶ **Plaining**

- ▶ **System Elevation**

- ▶ **Requirement Engineering**

- ▶ **Code**

199. Expert system technique Where a hypothesis is given at the beginning and the interface engine proceeds to ask the user questions about selected facts until the hypothesis is either confirmed or denied.

- ▶ **Backward chaining**

- ▶ **Data Mining**

- ▶ **Forward Chaining**

- ▶ **Network Knowledge**

200. Reasoning is backward chaining is known as:

- ▶ **Intelligence-Driven reasoning**

- ▶ **Goal-Driven reasoning**

- ▶ **Rule-Driven Reasoning**

- ▶ **Data-Driven Reasoning**

201. Backward chaining is more focused and tries to avoid exploring _____ Paths of reasoning.

- ▶ **Un-Necessary**

- ▶ **Different**

- ▶ **Necessary**

- ▶ **Searching**

202. _____ is the part of the system that controls the process of deriving conclusions.

- ▶ **reasoning or inference**